AI Knowledge Base for Claude Code: case study

Built by Mario Chasco, October 2026.

What it is

A local, auto-updating AI knowledge base with an MCP server and a per-prompt router for Claude Code. It keeps official docs, changelogs and creator content searchable, and injects only the one or two most relevant pointers into each prompt instead of dumping context.

Architecture

```
31 sources --> daily sync (08:00) --> transcripts / notes written by Claude
   --> index + MCP tools: kb_search / kb_get / kb_whats_new
   --> ROUTER.md + UserPromptSubmit hook ([ROUTER] line per prompt)
   --> weekly promotion branch --> Markdown AI OS (Obsidian vault)
```

Numbers

- 31 sources: canonical docs, changelogs, GitHub releases and 15+ creator channels.
- Hook cost: about 0.25 s and at most 1200 characters per prompt.
- Router precision: first hit 0.44 to 0.70; correct within two hits 0.80; 0 false positives on short prompts.
- Retry backoff for failed fetches: 1 h / 6 h / 24 h / 48 h.

What broke and how it was fixed

- **YouTube IP block:** added a proxy check, exponential retry backoff and a `kb doctor` health command.
- **Low router precision:** added synonyms, ES/EN stemming, IDF weights and top-2 results; measured against a fixed test set.
- **Mac Python 3.9 incompatibility:** made the hook code compatible with older Python so the same hooks run on both machines.

Stack

Python, Claude Code with subagents, Anthropic API, yt-dlp, rclone, systemd, Markdown/Obsidian, git.